



## “Weekly Meeting Notes”

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## Week 1 – Project Kickoff & Team Contract

*Date: 09 August 2025*

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- All team members reviewed and signed the Team Contract with assigned roles:
  - ✓ Mansour Saeed – Project Lead (Oversees Agile-Spiral oversight, stakeholder coordination).
  - ✓ Ali Alhammadi – Technical Lead (Focuses on NIST/GDPR compliance, Docker deployment).
  - ✓ Abdulla Almessabi – QA & Communications Lead ( responsible for SFTP/TLS 1.3 implementation, unit testing).
- Established communication protocols: Slack for daily standups, Jira for task tracking, and Git for version control.
- Agreed on PEP8 standards, 48hour peer review windows, and biweekly sprint meetings.
- Outlined initial goals: Modular agent design, 60% efficiency improvement, GDPR compliance.

| Name                     | Attendance |
|--------------------------|------------|
| Mansour Saeed Alhamadani | Attended   |
| Ali Alhammadi            | Attended   |
| Abdulla Almessabi        | Attended   |

## Week 2 – System Requirements & Architecture Planning

*Date: 13 August 2025*

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Ali presented core libraries:

- `hashlib` for SHA256 hashing (NISTcertified).
- `PyPDF2` and `ExifTool` for metadata extraction.
- `paramiko` for SFTP/TLS 1.3 communication.

Abdulla proposed SQLite for the blackboard database and Docker for crossplatform deployment.

Mansour drafted the Agile-Spiral timeline:

- Sprint 1: Search Agent development (file signature analysis).
- Sprint 2: Processing Agent (hashing, metadata).
- Sprint 3: Archiving/Communication Agent integration.
- Begun drafting System Requirements section.

| Name                     | Attendance |
|--------------------------|------------|
| Mansour Saeed Alhamadani | Attended   |
| Ali Alhammadi            | Attended   |
| Abdulla Almessabi        | Attended   |

## Week 3 – Agent Design & Technical Implementation

*Date: 17 August 2025*

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Ali prototyped the Search Agent:

- Implemented binary header analysis (50+ file types, e.g., `0x25504446` for PDFs).
- Reduced false positives by 37% in testing.

Abdulla configured Docker containers for Windows/Linux compatibility.

Mansour finalized UML class diagram (inheritance hierarchy for agents).

Team debated Python vs. C++:

- Chose Python for forensic library integration (The Sleuth Kit API) and rapid prototyping.
- Drafted Design Decisions section (blackboard architecture, AES256 encryption).

| Name                     | Attendance |
|--------------------------|------------|
| Mansour Saeed Alhamadani | Attended   |
| Ali Alhammadi            | Attended   |
| Abdulla Almessabi        | Attended   |

## Week 4 – Agile-Spiral Risk Mitigation & Testing

*Date: 24 August 2025*

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- Conducted risk assessment phase:
- Prioritized writeblocker emulation (`os.O\_RDONLY` flag) to prevent tampering.
- Tested threadsafe database writes with `concurrent.futures`.
- Abdulla implemented SFTP retries (exponential backoff: 1s, 2s, 4s).
- Ali validated SHA256 outputs against NIST test vectors (e.g., “abc” → `ba7816bf...`).
- Begun integration testing (10,000 files processed, 60% speed improvement).

| Name                     | Attendance |
|--------------------------|------------|
| Mansour Saeed Alhamadani | Attended   |
| Ali Alhammadi            | Attended   |
| Abdulla Almessabi        | Attended   |

## Week 5 – Challenges & Graphical Design

*Date: 31 August 2025*

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Addressed scalability:

- Ali optimized multithreading (3.1 hours for 1 TB data).
- Abdulla resolved GDPR compliance via RBAC and AES256 encryption in Archiving Agent.
- Mansour finalized sequence/activity diagrams (Appendices B, C):
- Workflow from search initiation to SFTP transfer.
- Error handling for EXIF extraction failures.
- Drafted Challenges and Solutions section.

| Name                     | Attendance |
|--------------------------|------------|
| Mansour Saeed Alhamadani | Attended   |
| Ali Alhammadi            | Attended   |
| Abdulla Almessabi        | Attended   |

## Week 6 – Report Drafting & Final Review

*Date: 07 September 2025*

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- Mansour consolidated Academic Grounding (Wooldridge’s multiagent theory, Carrier’s forensic guidelines).
- Ali reviewed encryption protocols (TLS 1.3, RFC 8446 compliance).
- Abdulla formatted References (APA) and crosschecked NIST/GDPR citations.
- Team performed code review:
- Validated threadsafe operations and GDPRcompliant logging.
- Ensured 100% peerreviewed pull requests.
- Prepared presentation slides on courtadmissible results and 60% efficiency gains.

| Name                     | Attendance |
|--------------------------|------------|
| Mansour Saeed Alhamadani | Attended   |
| Ali Alhammadi            | Attended   |
| Abdulla Almessabi        | Attended   |

## **Week 7 – Submission & Presentation**

***Date: 07 September 2025***

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- Submitted final report and codebase to portal.
- Conducted dry run of presentation:
- Highlighted blackboard architecture, AgileSpiral methodology, and legal compliance.
- Abdulla archived Docker images and SFTP keys securely.
- Celebrated project completion with a team dinner!

| <b>Name</b>              | <b>Attendance</b> |
|--------------------------|-------------------|
| Mansour Saeed Alhamadani | Attended          |
| Ali Alhammadi            | Attended          |
| Abdulla Almessabi        | Attended          |